Macro Transmission Model (VARX Framework)

Executive Summary

Objective

To quantify dynamic relationships between equity, foreign exchange, and global interest rates using a data-driven time-series framework.

Model Framework

A **Vector Autoregression with Exogenous Variable (VARX)** was implemented with:

- **Endogenous Variables**
  - Nifty 50
  - USD/INR
  - US 10Y Treasury Yield
- **Exogenous Variable**
  - Brent Crude Oil

Data was transformed using **ADF-based stationarity checks**, with log-differencing applied where required.

Key Findings

1. USD/INR Dynamics

- Strongly influenced by:
  - Brent crude (highly significant coefficient)
  - Lagged US 10Y yield movements
- Indicates:

INR is highly sensitive to global commodity and rate shocks

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2. Nifty 50 (Equity Market)

- Influenced by:
    - Oil shocks (statistically significant)
    - FX movements (lagged USD/INR effects)
- Suggests:

Equity markets absorb macro shocks through currency and commodity channels

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3. US 10Y Yield

- Highly autoregressive (self-driven)
- Weak dependence on oil
- Confirms:

Global rates act as a macro anchor rather than a reactive variable

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4. Impulse Response Analysis

- Oil shocks impact FX faster than equities
- Yield shocks transmit gradually across assets
- Effects are short-lived but economically meaningful

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5. Variance Decomposition (FEVD)

- Variables are largely self-driven in short horizons
- Over time:
    - FX variance increasingly influenced by oil and yields
    - Equity partially explained by FX shocks

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6. Risk Monitoring Insights

- Rolling correlation:

    - Mostly negative (typical EM pattern)
    - Temporary breakdowns observed
- Rolling beta:
    - Indicates changing sensitivity of equities to FX

Limitations

- Linear model (VAR) — does not capture nonlinear dynamics
- No structural break or regime-switch modeling
- Brent crude used as single exogenous variable
- Forecast assumes constant external conditions

Actionable Insights

📊 **Portfolio Strategy**

- Rising oil + rising yields → increased downside risk for equities

🛡️ **Risk Management**

- Monitor correlation breakdowns → hedge inefficiency risk

🌍 **Macro Positioning**

- US 10Y yield is key leading indicator
- Oil shocks impact FX before equities

Conclusion

The VARX framework successfully captures cross-asset transmission dynamics in an emerging market setting and provides a structured foundation for macro analysis and risk monitoring.